

Redesign of a Broken Healthcare Workflow Using AI & Automation

AI-Driven Transformation of the Pre-Authorization Workflow

GitHub Link: <https://github.com/rutvikbarbhai/AI-Powered-Healthcare-Operations-Design-Automation-Using-n8n>

Target Persona: Insurance Verification & Pre-Authorization

Team Focus Area: Revenue Cycle Management (RCM) Optimization

1. Executive Summary

During a full-day hospital site visit, the Pre-Authorization process was identified as a critical bottleneck impacting both patient care delivery and hospital revenue realization. The current workflow is manual, reactive, and highly dependent on individual effort. This proposal outlines an AI-agentic redesign that automates clinical data extraction and payer interactions, significantly reducing administrative workload while accelerating time-to-care.

2. Current State of Healthcare Workflow

The observed end-to-end workflow for a Pre-Authorization specialist is as follows:

- Trigger: A physician orders a high-cost procedure in the EHR.
- Portal Hunting: Specialist logs into payer portals to check authorization requirements.
- Data Scavenging: 20–30 minutes spent reviewing unstructured clinical notes for medical necessity evidence.
- Submission: Manual data entry or fax-based submission to insurers.
- Manual Follow-up: Excel-based tracking with periodic phone calls.
- Data Entry: Authorization ID manually updated in billing systems.

3. Redesigned Workflow:

The redesigned workflow leverages n8n as the orchestration layer and Large Language Models (LLMs) for clinical reasoning and document synthesis. Automation is introduced without disrupting existing clinical systems.

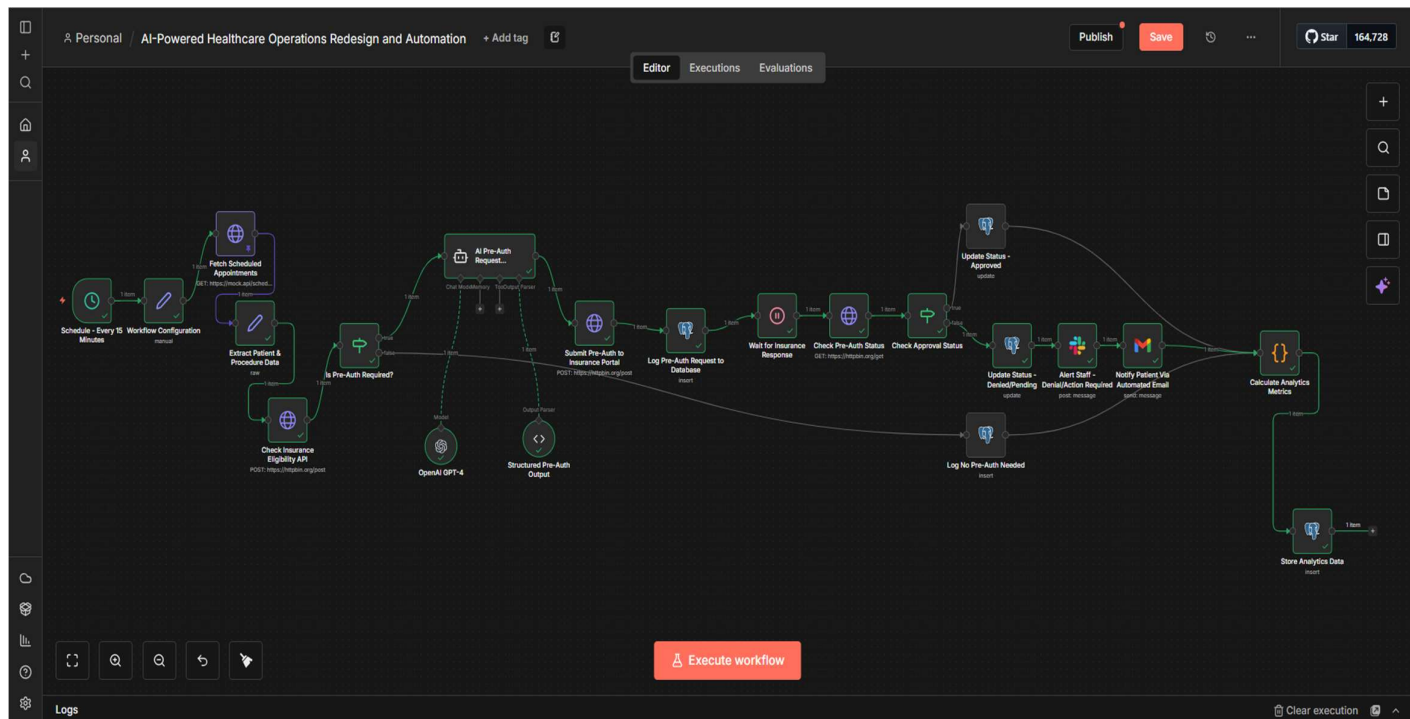
AI-Automated Flow:

- Step 1: Appointment Schedule Review
- Step 2: Patient & Procedure Data Collection
- Step 3: Insurance Eligibility Verification
- Step 4: Pre-Authorization Requirement Determination
- Step 5: Clinical Justification Writing
- Step 6: Pre-Authorization Submission
- Step 7: Logging & Tracking
- Step 8: Waiting & Follow-Ups
- Step 9: Approval / Denial Handling
- Step 10: Patient Communication
- Step 11: Analytics & Insights

4. Assumptions Made

- Hospitals have access to scheduling and insurance APIs (or exports)
- Insurance rules can be encoded at a basic level
- AI outputs are **reviewable and auditable**
- Automation is advisory + operational, not clinical decision-making
- Mocked systems are used for demonstration

Workflow Visualization



5. Operational Assumptions & Requirements

Data Inputs: EHR access via FHIR APIs, payer policy manuals (PDF), 270/271 eligibility data.

Assumptions: Digital-first infrastructure, structured clinical documentation, and permission for API/RPA-based integration.

1. Expected Real-World Impact

Metric	Current State	Redesigned State
Processing Time	45–60 mins/case	5–8 mins/case
Turnaround Time	5–7 days	< 24 hours
Error Rate	≈12%	< 2%
Cost per Authorization	~\$13.00	~\$2.50

2. High-ROI Insights Unlocked

- Payer Friction Index for contract optimization
- Documentation gap analysis by physician
- Pre-submission denial prediction
- Revenue leakage and patient drop-off tracking
- Labor allocation optimization toward high-value appeals

3. Technical Implementation (n8n & GitHub)

The prototype is implemented using n8n with modular nodes for clinical extraction, policy retrieval, and payer interaction. The GitHub repository contains the n8n workflow JSON, OpenAI nodes, Postgres policy database, and HTTP/RPA integration logic.

4. AI Usage Disclosure

AI tools were used to structure the analysis and articulate findings clearly. All operational assumptions and redesign decisions are based on real-world hospital workflow observations